Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption
One potential explanation for the website's connection timeout error message is: The website is experiencing a **Direct Denial of Service (DoS) attack**, specifically a **SYN flood attack**. **The logs show that:** A single IP address (203.0.113.0) is sending an abnormally high volume of TCP [SYN] packets to the web server (192.0.2.1) on port 443. The server attempts to respond, but the volume of requests is too high, causing the server to run out of resources to handle legitimate connections. **This event could be:** A SYN flood attack. This is a type of DoS attack that exploits the TCP three-way handshake process by overwhelming the server with connection requests that are never completed.

Section 2: Explain how the attack is causing the website to malfunction
When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake: 1. The client sends a [SYN] (synchronize) packet to the server to request a connection. 2. The server responds with a [SYN, ACK] (synchronize-acknowledge) packet, allocating resources for the connection. 3. The client sends an [ACK] (acknowledge) packet back to the server, completing the connection. **Explain what happens when a malicious actor sends a large number**

of SYN packets all at once:
The server allocates system resources (memory and processing power) for each [SYN, ACK] it sends back. Because the attacker never sends the final [ACK] to complete the handshake, these connections remain "half-open." The server's connection table fills up, exhausting its capacity to accept new, legitimate connections.

Explain what the logs indicate and how that affects the server:
The logs show the attacker's IP (203.0.113.0) flooding the server with [SYN] packets. As the attack progresses, the server becomes overwhelmed and stops responding to legitimate employee IPs (like 198.51.100.x). The logs show legitimate connection attempts failing with [RST, ACK] (reset) packets and HTTP/1.1 504 Gateway Time-out errors. Eventually, the server stops responding to anyone except the attacker.